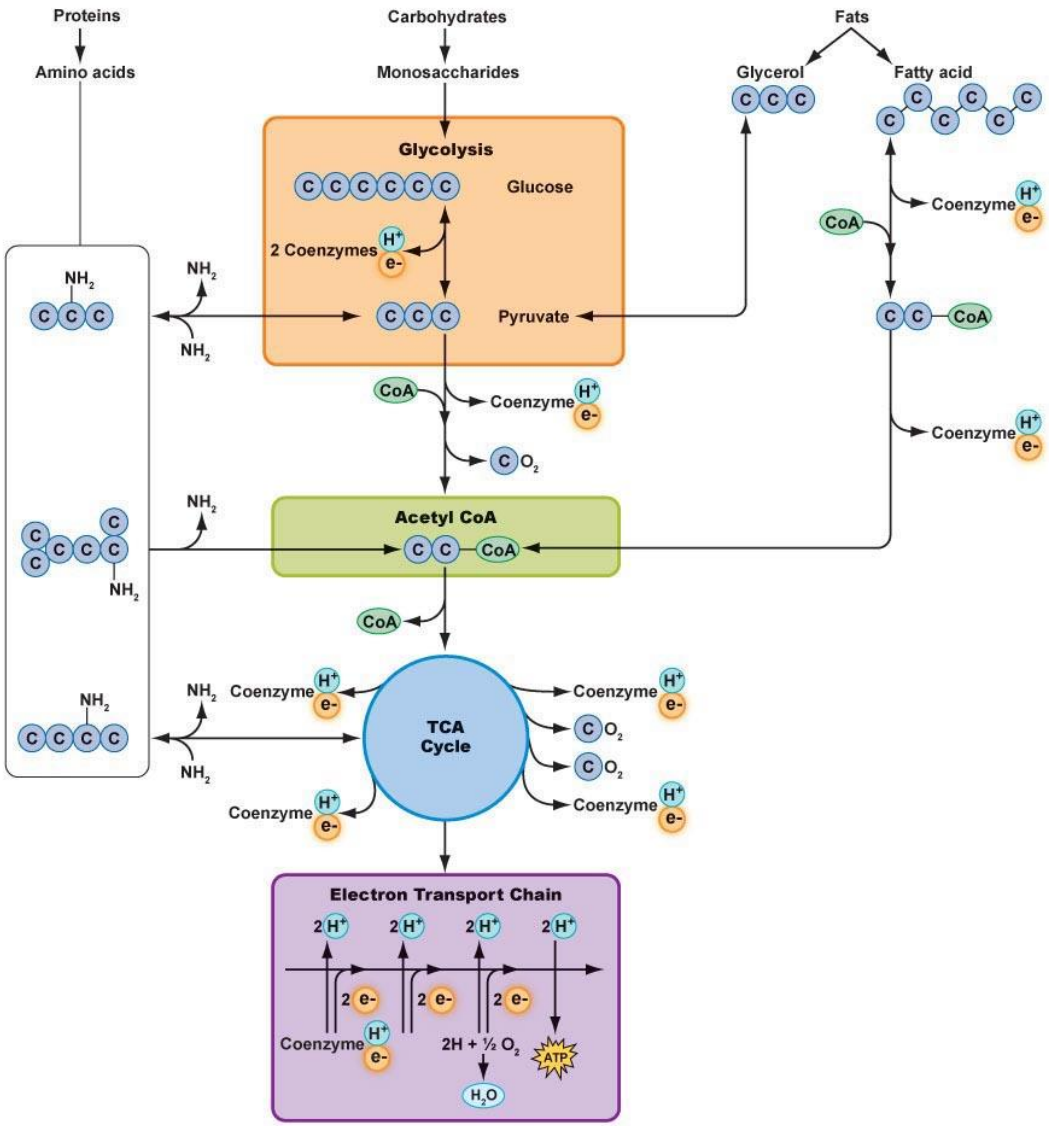
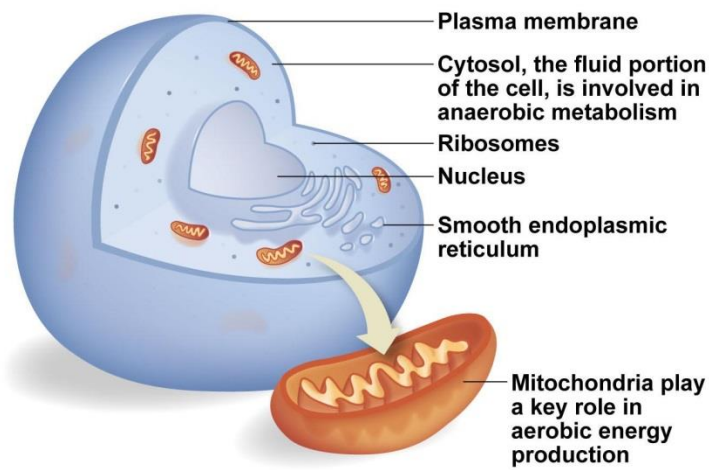
Carbohydrate Metabolism

Ram Sagar Misra

Your Body's Metabolism





- Sum of all chemical reactions in the body's cells
  - Generation of energy from carbohydrates, proteins, and fats
    - Anaerobically
    - Aerobically
  - Production of biological compounds
    - Nonessential amino acids
    - Intermediate substances needed for metabolism

- Never stops
- Adapts to individual needs and the environment
- Has several metabolic pathways
  - Glycolysis
  - TCA cycle
  - Electron transport chain

Overview of Energy Metabolism

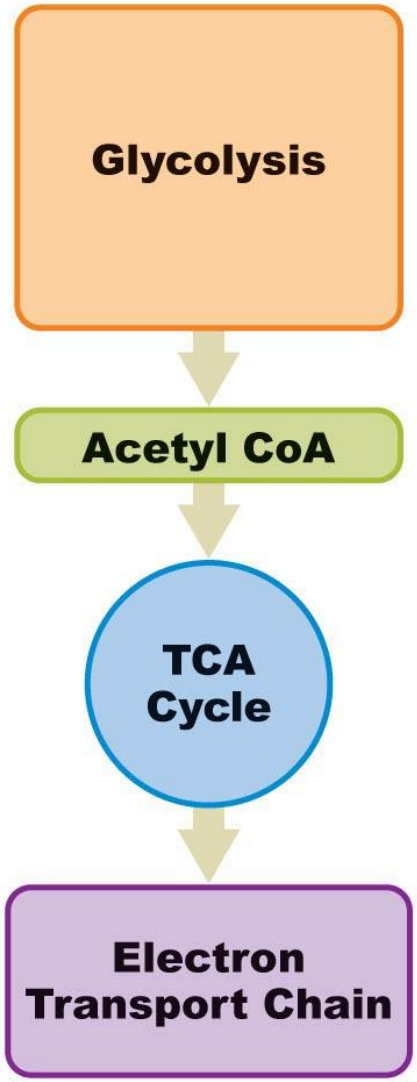
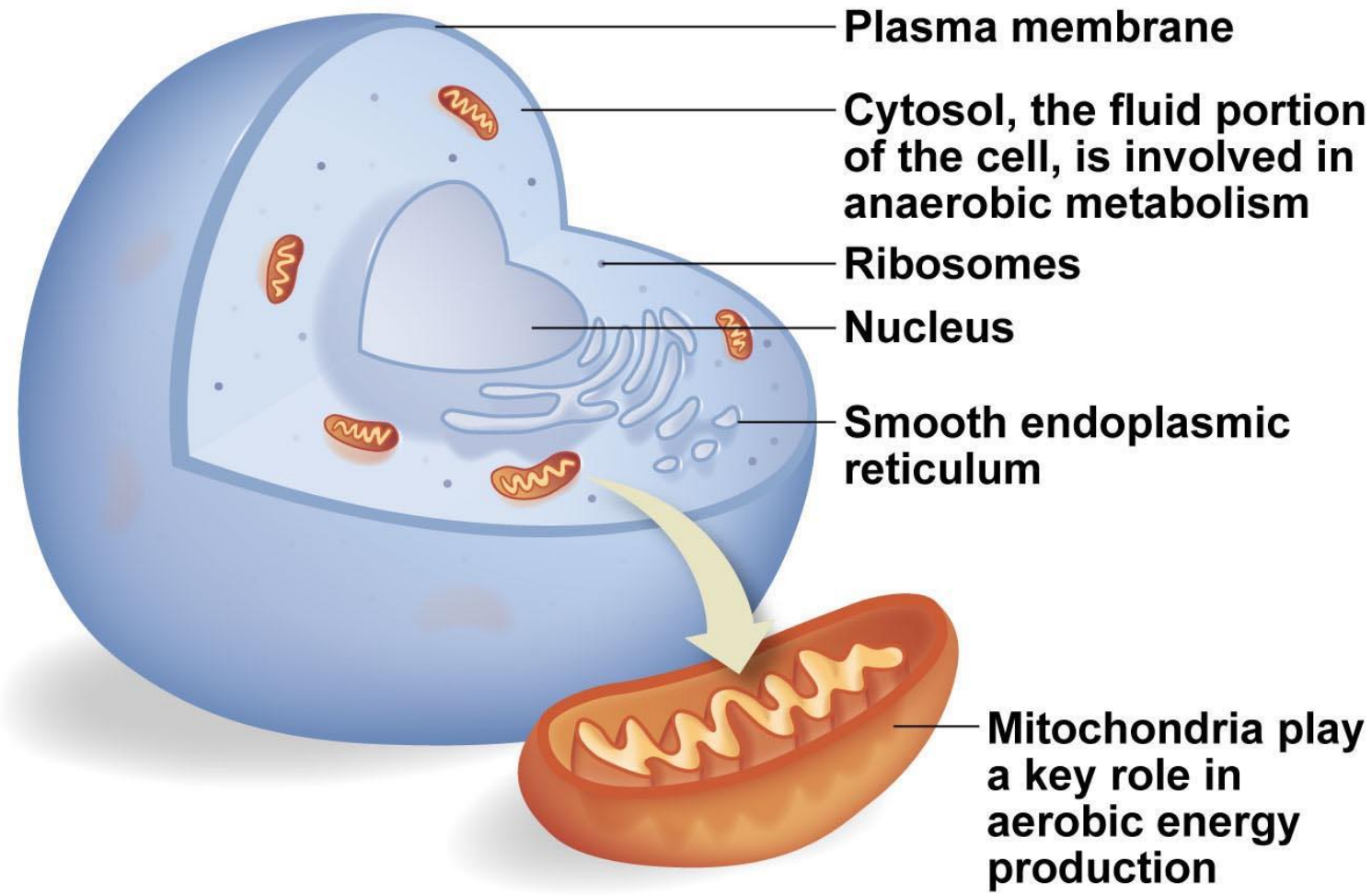


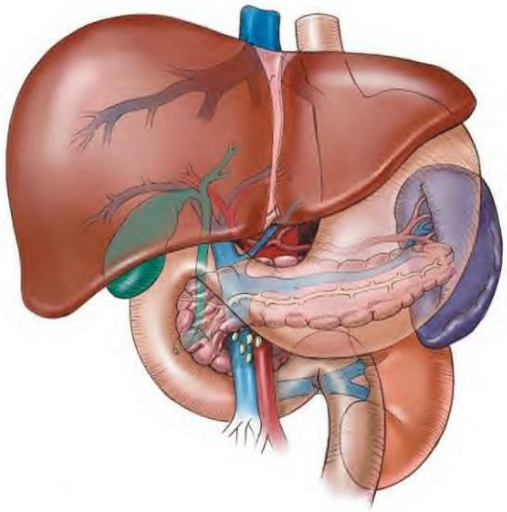
Figure 8.1

Metabolism Takes Place within Cells



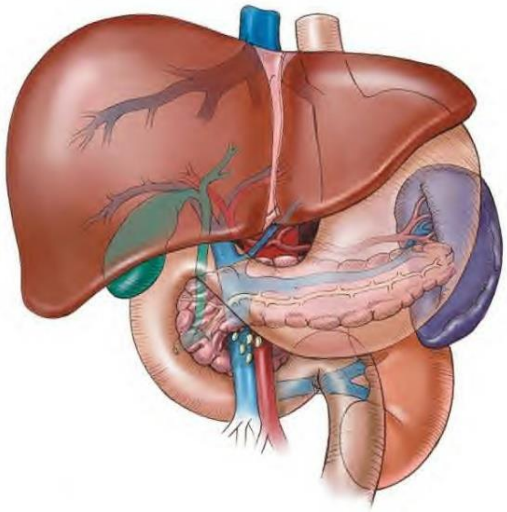
Liver

- Is the most metabolically active organ in the body
- First organ to metabolize, store, and distribute nutrients after absorption
- Proteins, carbohydrates, and fats are absorbed as:
  - Amino acids
  - Monosaccharides
  - Glycerol and fatty acids

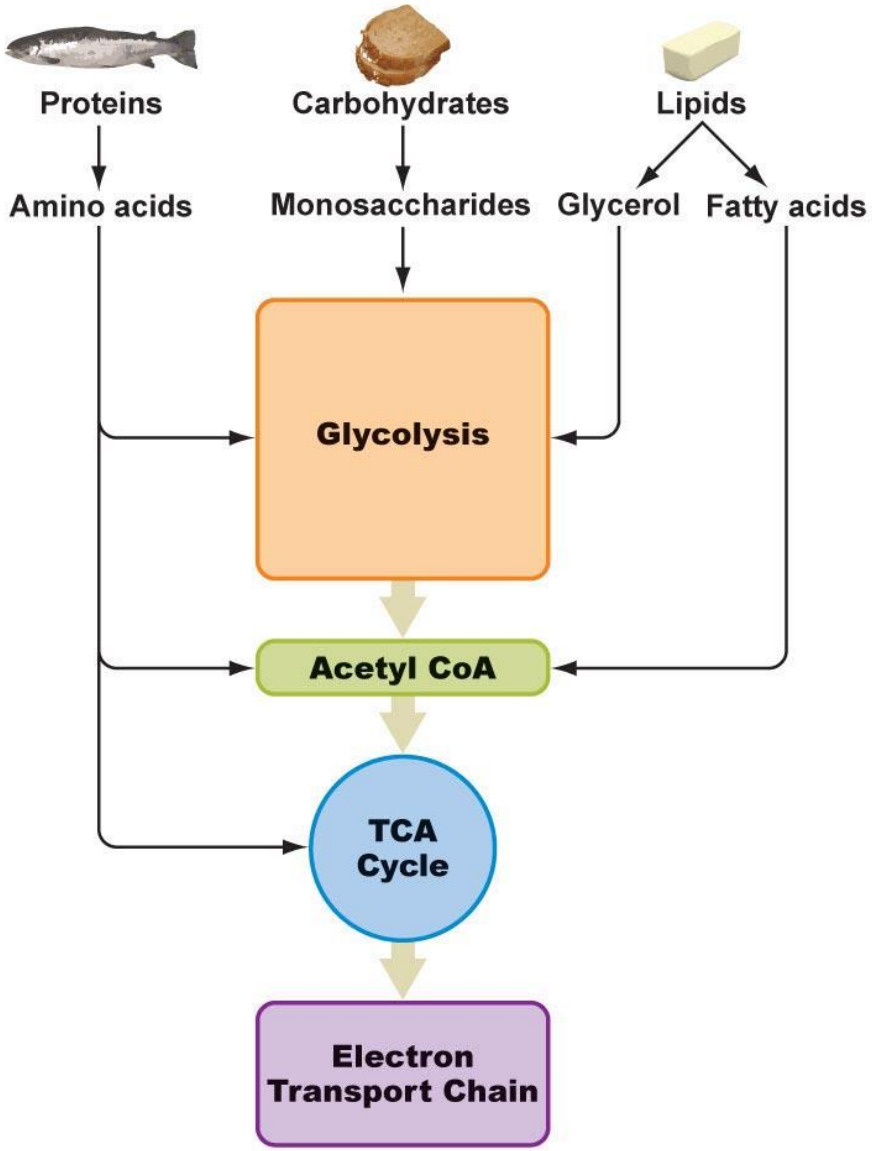


Liver

- Proteins, carbohydrates, and fats are in the liver converted to
  - Usable forms of energy
  - Storage forms
    - Glycogen
    - Triglycerides



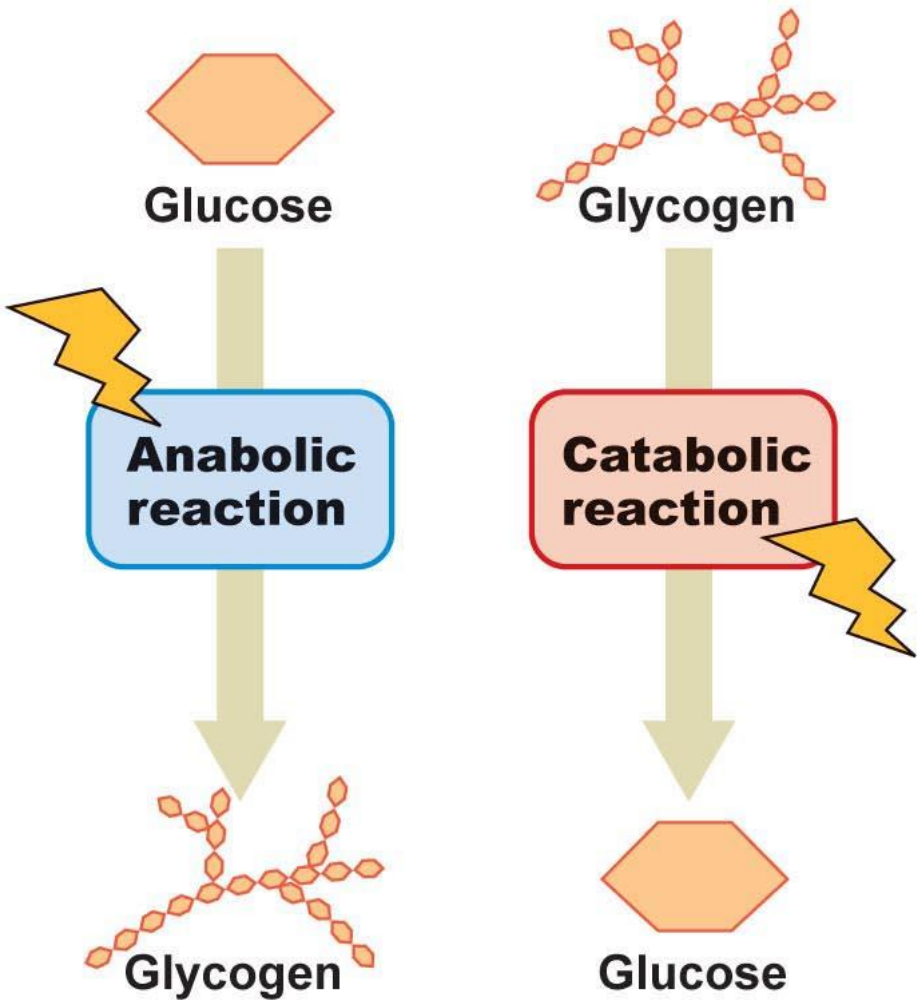
The Metabolic Fate of Food



Metabolic Pathways

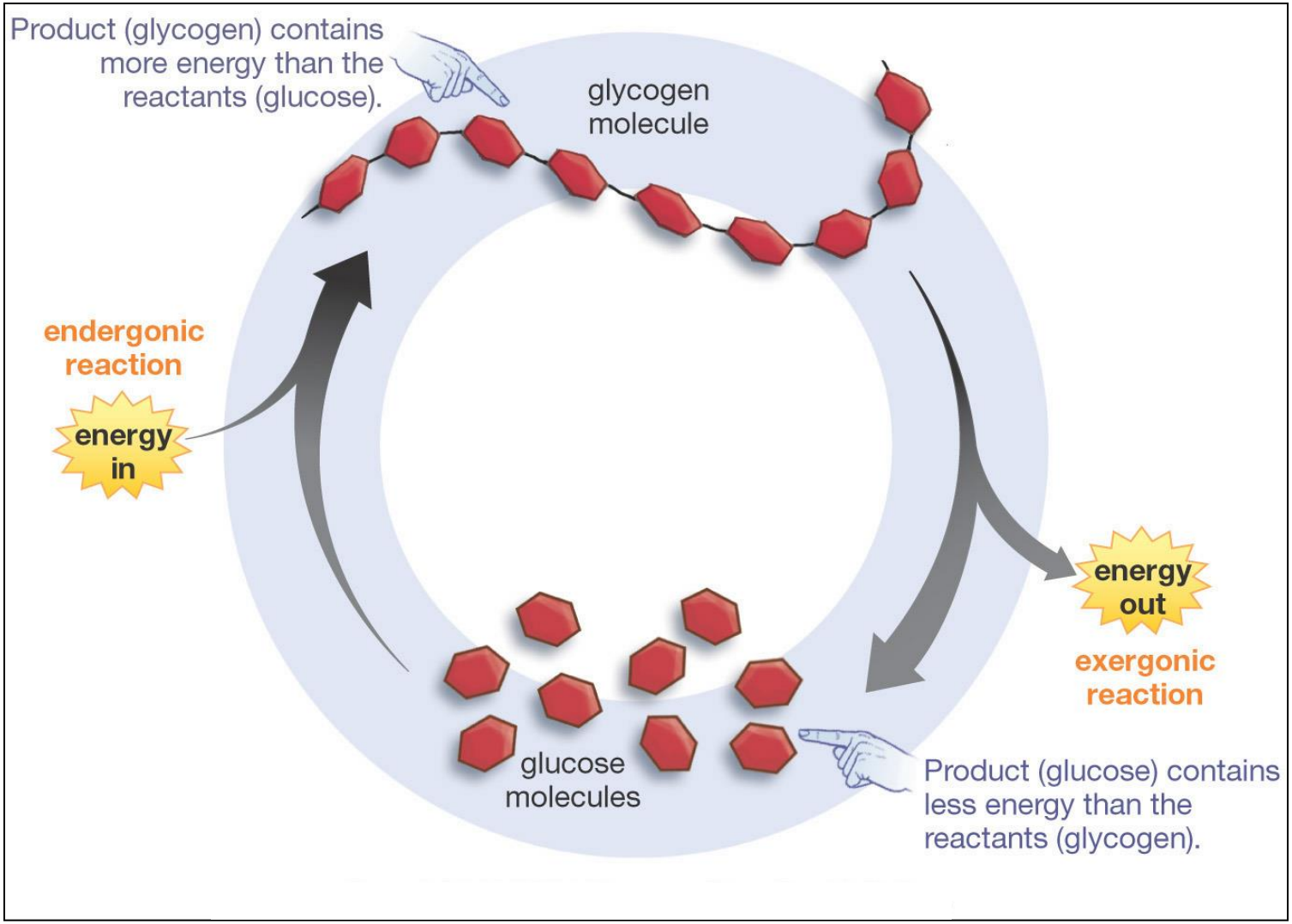
- A sequence of reactions that convert compounds from one form to another in the production of energy
- Different nutrients follow different pathways
- All pathways eventually converge into a pathway called the TCA cycle

Anabolic and Catabolic Reactions



a Anabolic reactions use ATP to combine smaller molecules, such as glucose, into larger compounds, such as glycogen, and release small amounts of heat.

b Catabolic reactions break down larger molecules into smaller compounds and release energy in the form of ATP and heat.

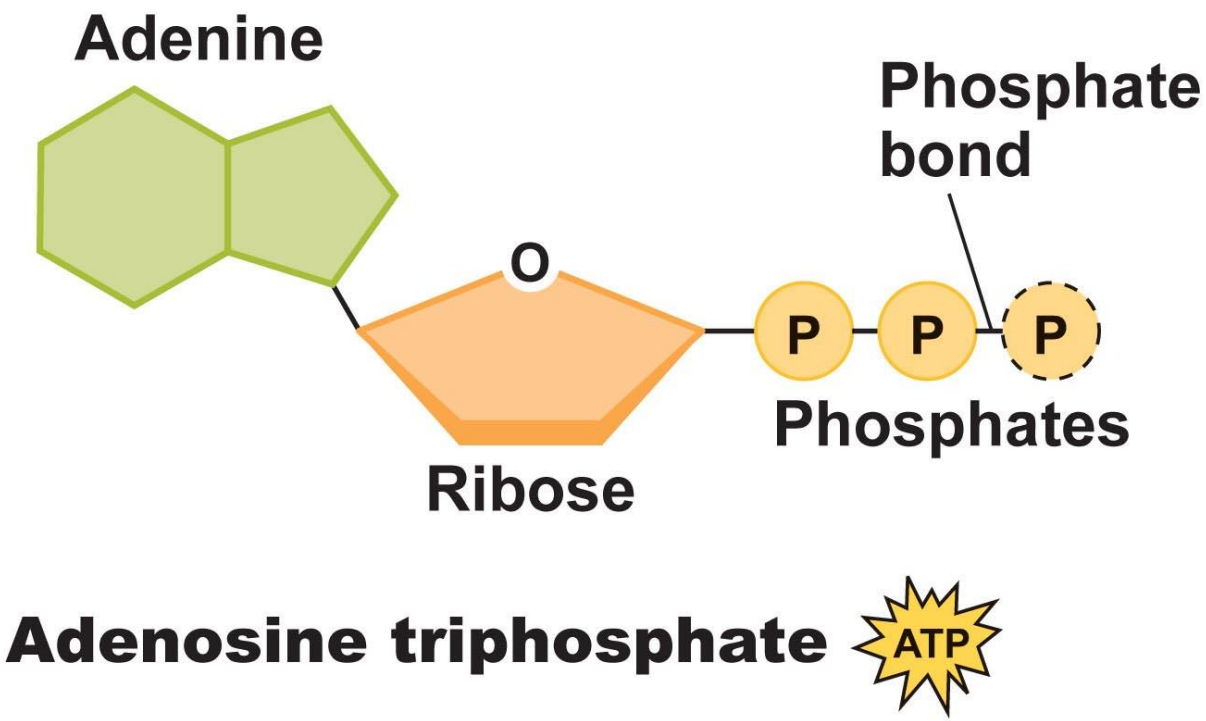

Product (glycogen) contains more energy than the reactants (glucose).
glycogen molecule
endergonic reaction
energy in
energy out
exergonic reaction
glucose molecules
Product (glucose) contains less energy than the reactants (glycogen).

Quick Review

- Metabolism is the sum of all metabolic processes that occur in the cells
- Mitochondria is the site of most of the metabolic reactions
- Metabolic processes follows specific pathways
  - Anabolic which use energy to build new substances
  - Catabolic which produce energy by breaking down molecules
- Enzymes and coenzymes catalyze reactions
- Hormones regulate reactions

Adenosine Triphosphate (ATP)

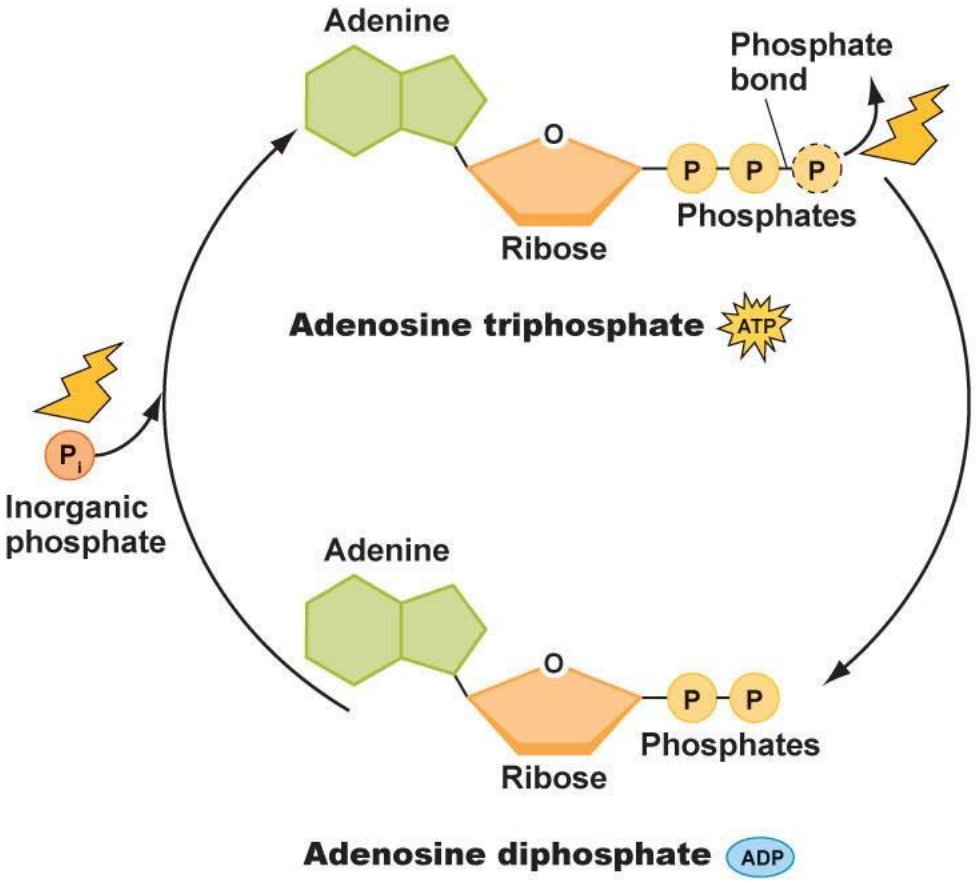
- A high-energy molecule composed of adenine, ribose, and three phosphate molecules



Adenosine Triphosphate (ATP)

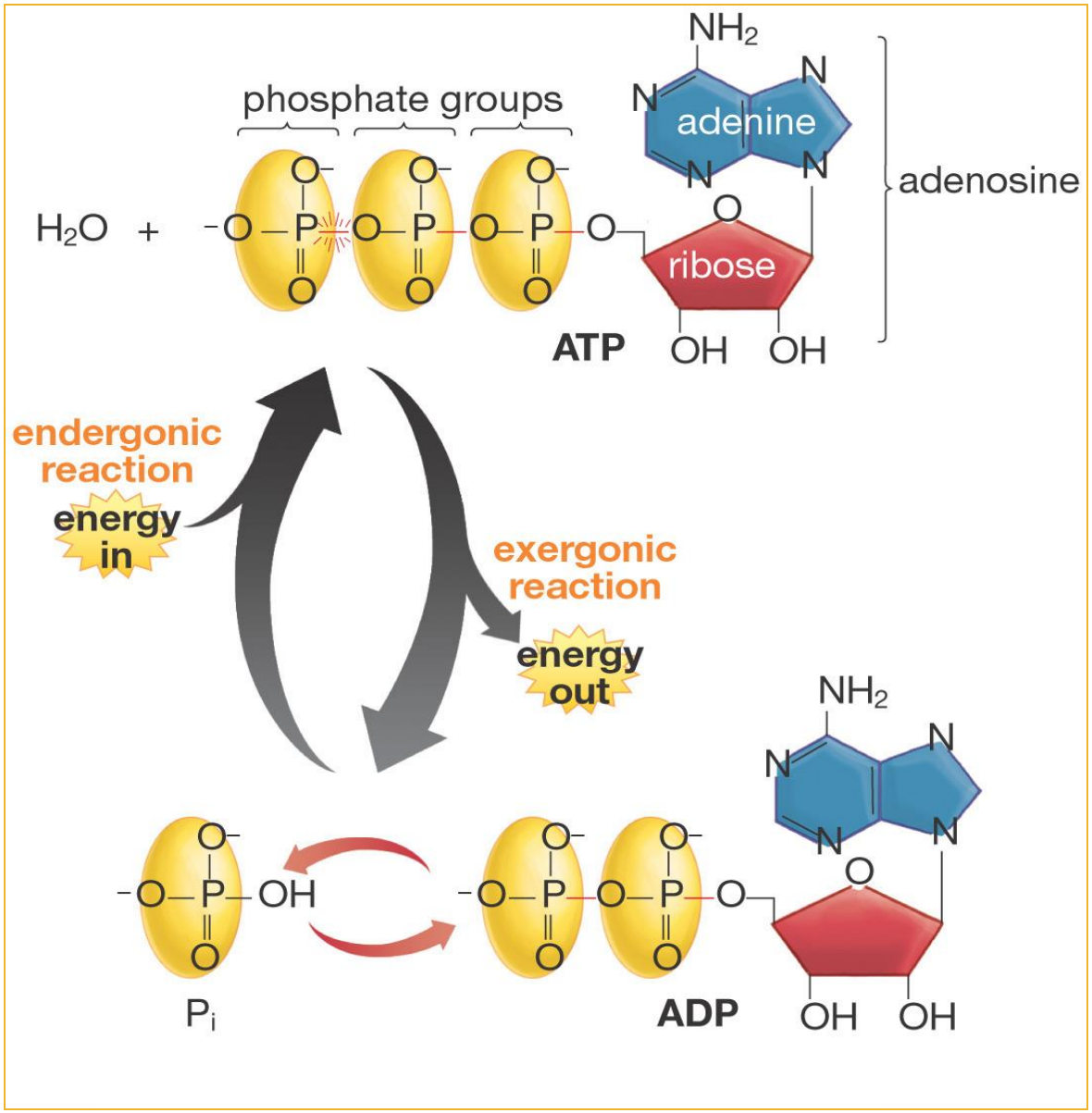
- Only source of energy used directly by the cell
- Energy is stored in the bonds that connect the phosphate groups
- The body must continually produce ATP to provide a constant supply of energy

ATP to ADP (Adenosine Diphosphate)



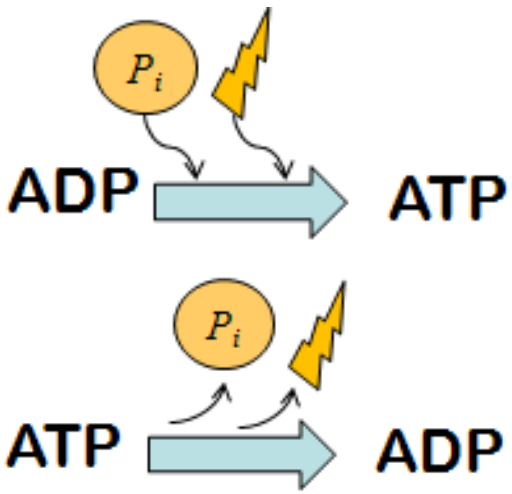
1 When you need energy, your body hydrolyzes one of the phosphate bonds, releasing one phosphate and a tremendous amount of energy.

2 With the release of one phosphate from ATP, a new molecule with only two phosphates is formed, adenosine diphosphate, or ADP. A phosphate group is added back to ADP to reform ATP during catabolism.


phosphate groups
adenine
adenosine
ribose
ATP
H_2O +
NH_2
OH
endergonic reaction
energy in
exergonic reaction
energy out
P_i
ADP

Creatine Phosphate

- High-energy compound in muscle cells
- Creatine combine plus inorganic phosphate
-



Creatine Phosphate

- Creatine monohydrate – a supplement sold in stores
  - Marketed to athletes to maximize PCr stores
  - Research
    - Increased performance of short-duration, high-intensity activities
    - Side effects

Carbohydrates to Energy

- Carbohydrate metabolism is the backbone of energy production
- Glucose
  - Important energy source for the brain and red blood cells
  - Generates energy anaerobically and aerobically
  - Transforms to energy via four metabolic pathways
    - Glycolysis
    - Intermediate reaction pyruvate to acetyl CoA
    - Tricarboxylic acid (TCA) cycle
    - Electron transport chain

Glycolysis

- Breakdown of glucose
- First step in forming ATP from glucose
- Takes place in the cytosol of the cell
- Ten-step catabolic process
- One six-molecule glucose process to two three-carbon molecules of pyruvate and two molecules of ATP

Glycolysis

- The other monosaccharides can also be used to produce ATP
  - Fructose enters glycolysis after going through seven metabolic steps
  - Galactose enters after going through four metabolic steps

Glycolysis Summary

- Ten-step process that produces
  - Two molecules of ATP
  - Two molecules of pyruvate
  - Two energized coenzyme molecules
  - Two hydrogen ions that are transported to the electron transport chain
  - Two water molecules

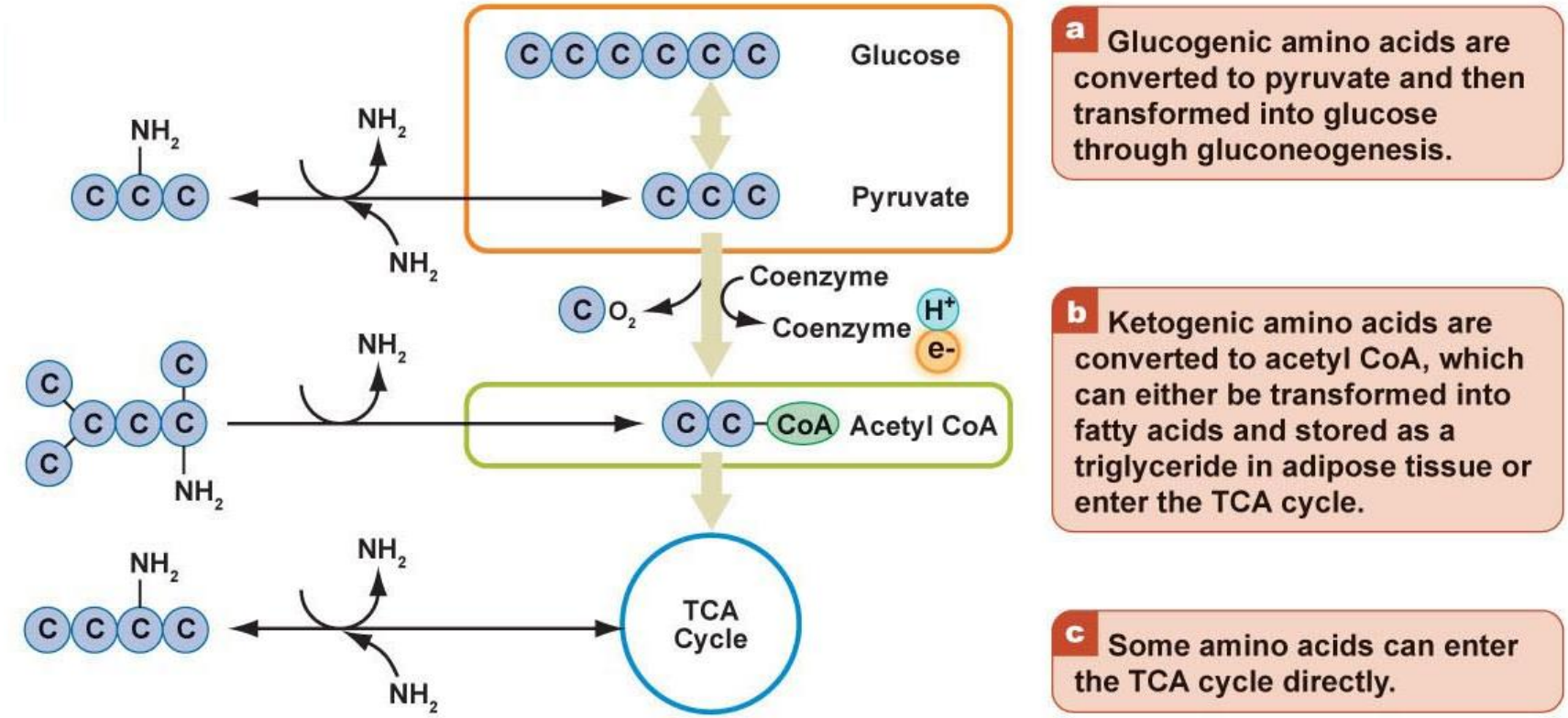
How Does Protein Provide Energy?

- Amino acids
  - Primary use/most important function is building protein
  - If the amino acid is not used to build protein the amine group must be removed through deamination
  - Excess can be used for energy production, converted to glucose, or stored as fat
  - Used, in a limited extent, for energy in diets low in kilocalories and/or carbohydrate

Ketogenic verses Glucogenic Amino Acids

- Ketogenic amino acids
  - Leucine
  - Lysine
- Both ketogenic and glucogenic amino acids
  - Isoleucine
  - Tryptophan
  - Phenylalanine
  - Tyrosine
- Glucogenic amino acids
  - The fourteen other amino acids

Glucogenic and Ketogenic Amino Acid Metabolism



<u>Acetyl CoA cannot be used to make glucose</u>

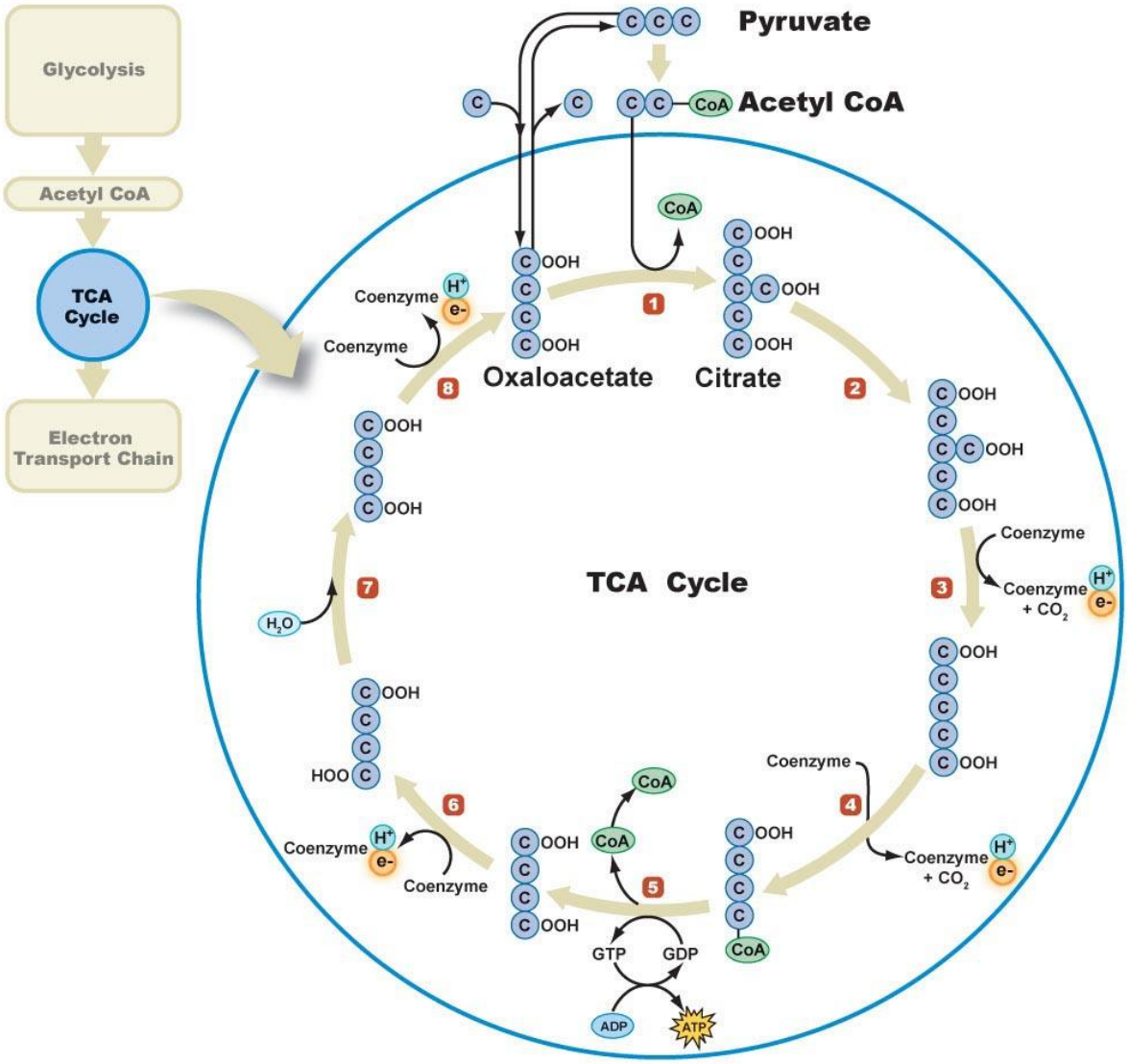
Amino Acids to Glucose

- Glucogenic amino acids
  - Major source of blood glucose when the diet is lacking in carbohydrate
  - Can come from food
  - Can come from the breakdown of muscle

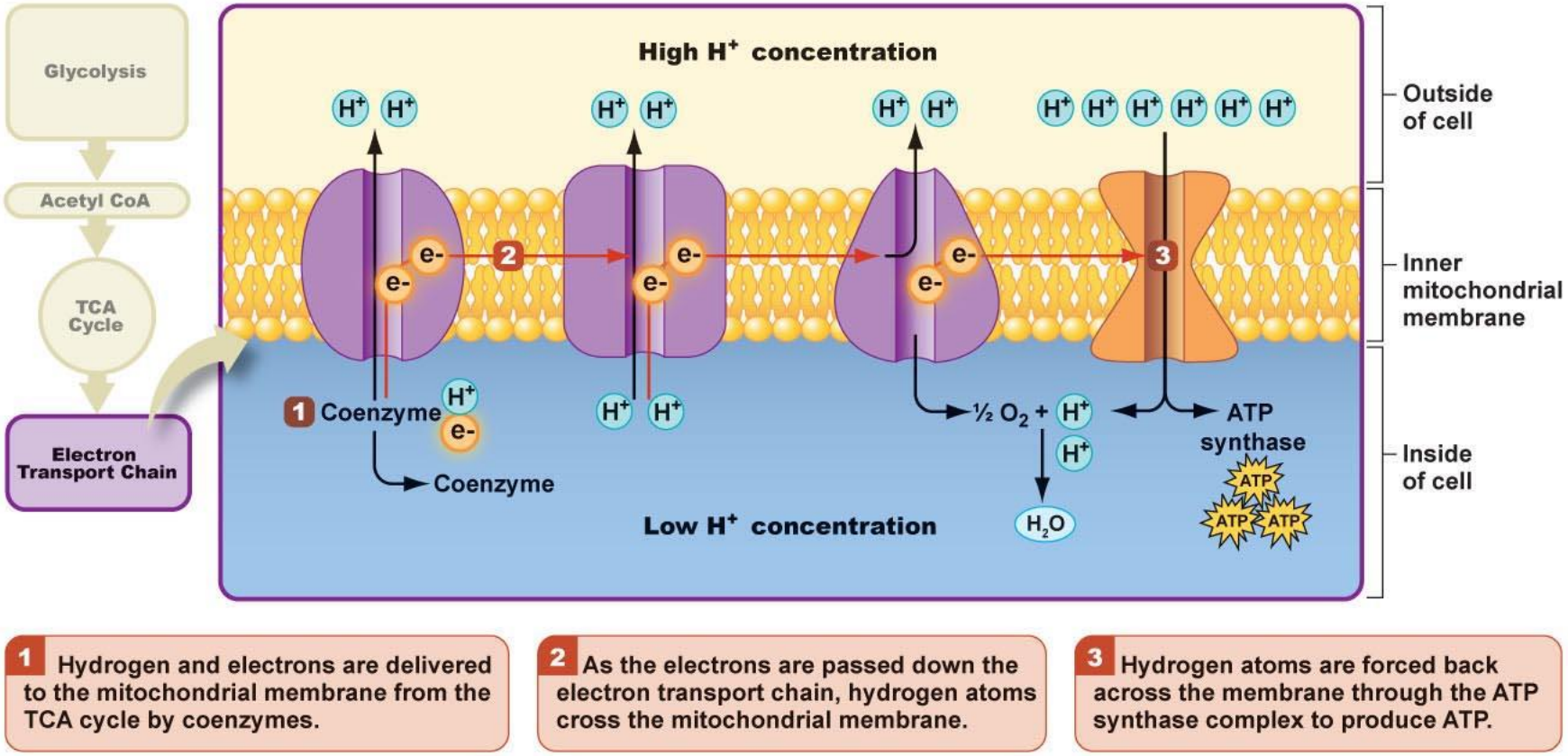
Quick Review

- Amino acids can
  - Be used to produce energy
  - Be used to produce glucose
  - Convert to fatty acids and can be stored as triglycerides
- Must be deaminated to be used for energy
- Once deaminated can be transformed into:
  - Pyruvate
  - Acetyle CoA
  - TCA cycle compounds
- Glucogenic amino acids can be converted to glucose

The TCA Cycle



The Electron Transport Chain



1 Hydrogen and electrons are delivered to the mitochondrial membrane from the TCA cycle by coenzymes.

2 As the electrons are passed down the electron transport chain, hydrogen atoms cross the mitochondrial membrane.

3 Hydrogen atoms are forced back across the membrane through the ATP synthase complex to produce ATP.

The Electron Transport Chain

- Flavoproteins
  - Protein complexes that transfer the electrons through the electron transport chain
  - Contain B vitamin riboflavin
- Cytochromes
  - Protein complexes that move electrons down the electron transport chain
  - Contain iron and copper
  - Iron deficiency can slow down this metabolic process

Though vitamins and minerals do not provide energy, they are essential for energy production

Overview of Energy Metabolism

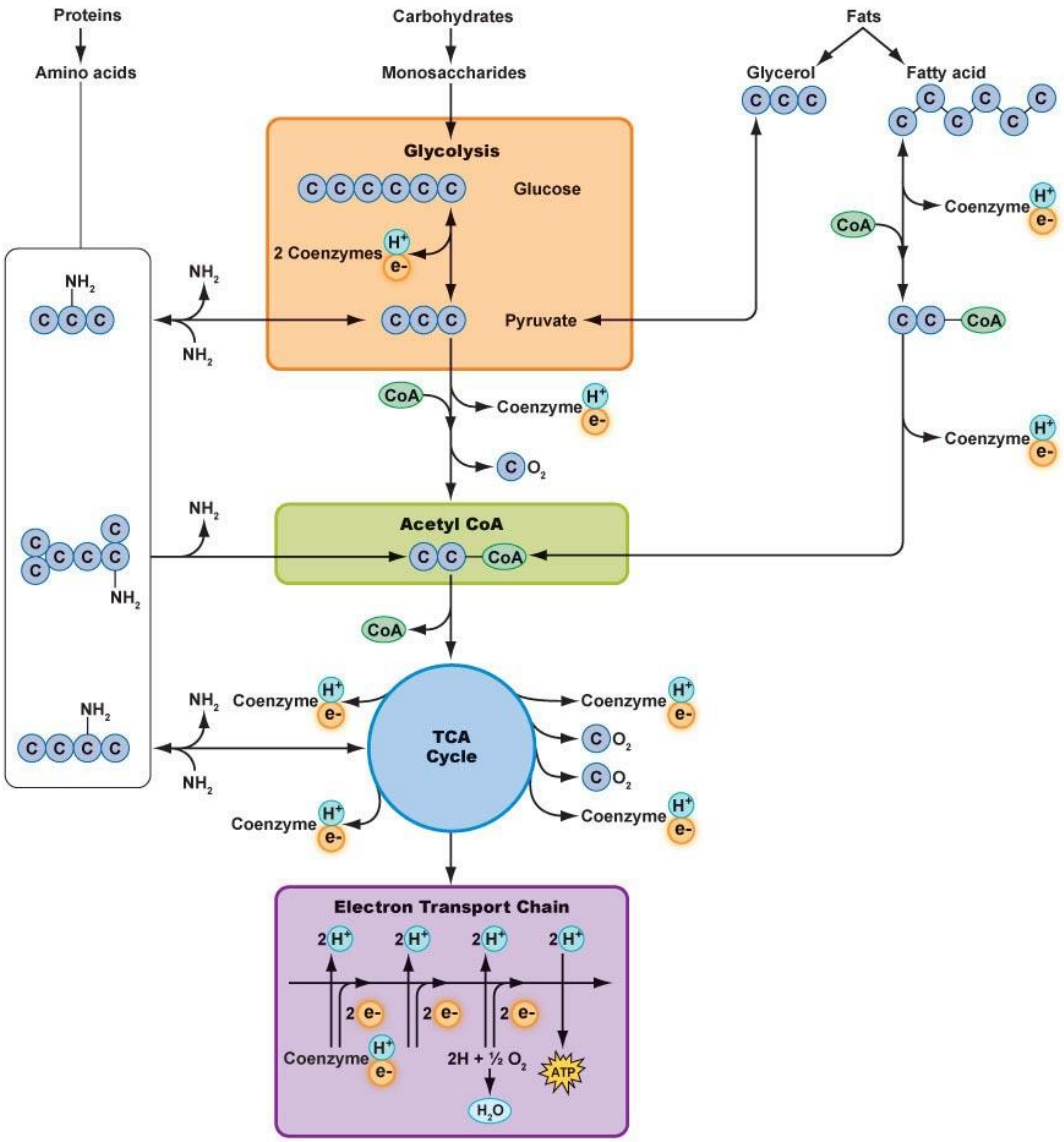
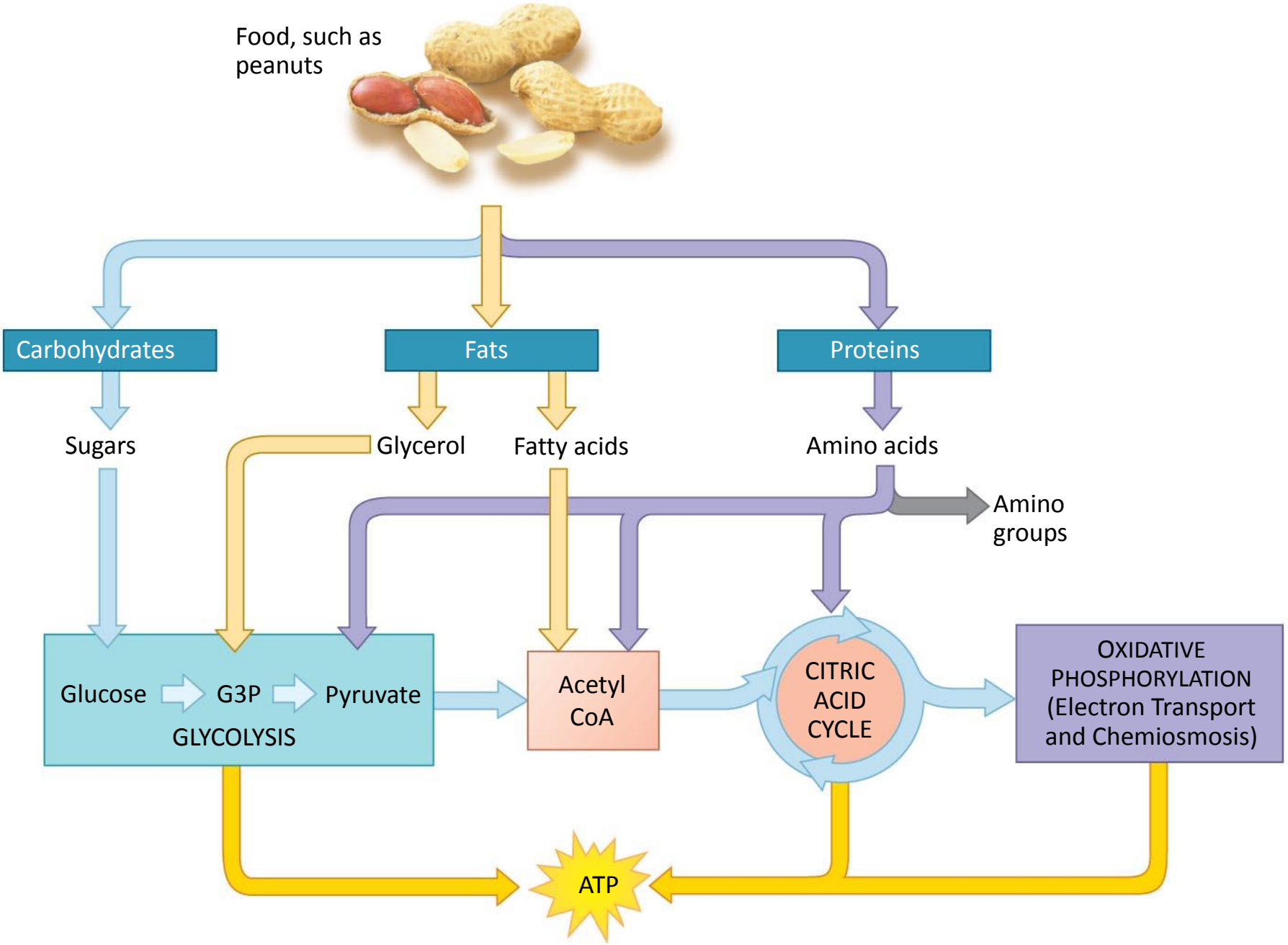


Figure 8.14


Food, such as peanuts
Carbohydrates
Fats
Proteins
Sugars
Glycerol
Fatty acids
Amino acids
Amino groups
Glucose
G3P
Pyruvate
GLYCOLYSIS
Acetyl CoA
CITRIC ACID CYCLE
OXIDATIVE PHOSPHORYLATION (Electron Transport and Chemiosmosis)
ATP